Explainable AI for Pleural Effusion Detection with Dual-View 3D Anatomical Localisation

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Abstract

Pleural effusion, the abnormal accumulation of fluid in the space surrounding the lungs, is one of the most common thoracic pathologies encountered in clinical practice, yet it remains a condition that is frequently difficult for medical students to understand spatially from standard two-dimensional chest radiographs alone.

This dissertation presents the design, implementation, and evaluation of an explainable AI pipeline for pleural effusion detection that bridges the gap between two-dimensional radiographic analysis and three-dimensional anatomical understanding. The system combines a unified DenseNet121 classifier trained on 224,316 chest radiographs from the Stanford CheXpert dataset with Grad-CAM++ explainability heatmaps, PSPNet lung segmentation corrected by a Statistical Shape Model, and a dual-view coordinate triangulation system that maps detected pathology locations onto an interactive three-dimensional lung model.

The classifier achieved 87.24% test accuracy and 94.11% AUC-ROC, exceeding the project targets of 85% accuracy and 0.93 AUC. Grad-CAM++ heatmap validation on 20 held-out cases produced anatomically correct costophrenic angle activation in 100% of True Positive cases. The Statistical Shape Model corrected PSPNet's systematic boundary truncation in effusion cases by a measured 31 pixels (13.8% of image height). The percentage-based 3D coordinate mapping correctly placed extracted coordinates at the inferior posterior region of the lung model, consistent with the known anatomy of gravity-dependent pleural fluid.

The key contribution of this project is a novel pipeline that connects dual-view Grad-CAM++ heatmaps from plain chest radiographs with a percentage-based coordinate mapping onto a three-dimensional anatomical lung model, specifically for educational visualisation of pleural effusion without requiring CT ground truth, camera calibration, or patient-specific geometry. This is supported by an SSM-based boundary correction strategy that addresses a specific and previously undocumented failure mode of parenchyma-trained segmentation models in the presence of pleural effusion.

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Introduction

Project Origins and Motivation

A recurring problem in medical education is the cognitive difficulty of translating two-dimensional imaging into three-dimensional anatomical understanding. A clinician interpreting a chest X-ray must mentally reconstruct the spatial relationships of structures that have been compressed into a flat projection. For an experienced radiologist this process becomes intuitive; however, for a student encountering radiographic interpretation for the first time, it represents one of the most challenging aspects of learning diagnostic medicine.

This challenge is particularly acute for pleural effusion, where the critical diagnostic feature, namely the blunting of the costophrenic angle caused by fluid accumulating inferiorly, is inherently spatial. Research on radiograph interpretation training consistently identifies spatial reasoning as the primary barrier to competency, and there is growing interest in using interactive three-dimensional visualisation tools to support this learning [Lessmann et al., 2016]. At the same time, deep learning systems for chest radiograph classification have matured to the point where their outputs carry genuine diagnostic weight, yet they remain largely opaque in terms of spatial reasoning. Connecting these two threads, specifically explainable AI outputs and anatomical 3D visualisation, is the motivating rationale for this project.

The scope was deliberately constrained by two technical principles: first, that reconstruction-based approaches, which attempt to generate 3D anatomy from 2D images using generative models, risk synthesising plausible but non-existent anatomical detail unsuitable for educational use; and second, that dual-view radiographs, which are already standard clinical practice, contain sufficient orthogonal information to support coordinate triangulation onto a reference 3D model without any additional data collection. These constraints led naturally to the design described in this dissertation.

Clinical Context and Motivation

Pleural effusion is the abnormal accumulation of fluid in the pleural space between the lung and chest wall. It is one of the most frequently encountered thoracic pathologies in clinical practice and serves as an important diagnostic indicator for underlying conditions including congestive heart failure, pneumonia, malignancy, and pulmonary embolism. Early and accurate detection of pleural effusion is essential for timely clinical intervention and improved patient outcomes.

Chest radiography remains the primary first-line imaging modality for detecting pleural effusion due to its widespread availability, low cost, and minimal radiation exposure compared to computed tomography (CT). However, interpreting chest X-rays presents significant challenges. Radiographic images are two-dimensional projections of three-dimensional anatomical structures, which means clinicians must mentally reconstruct spatial relationships from flat images. This cognitive demand is particularly difficult for medical students and junior doctors, who often struggle to correlate what they see on an X-ray with the actual anatomical locations in the body.

Furthermore, pleural effusion detection is complicated by the limitations of single-view imaging. Standard posteroanterior (PA) chest radiographs can miss small effusions because fluid settles in gravity-dependent posterior regions that are obscured by overlapping structures in the frontal view. Lateral radiographs are significantly more sensitive by visualising the posterior costophrenic sulcus. Despite this diagnostic advantage, lateral views are frequently underutilised in both clinical workflows and automated detection systems, representing a missed opportunity for improved diagnostic accuracy.

The Role of Artificial Intelligence in Medical Imaging

The application of deep learning to medical image analysis has demonstrated remarkable success, with convolutional neural networks (CNNs) achieving radiologist-level performance on numerous diagnostic tasks. For chest radiography specifically, architectures such as DenseNet and ResNet have been successfully deployed for multi-pathology classification, often achieving performance comparable to or surpassing human baselines on large-scale datasets.

However, the adoption of AI systems in clinical practice faces a critical barrier, commonly referred to as the "black box" problem. Deep learning models typically provide predictions without transparent reasoning, making it difficult for clinicians to understand why a particular diagnosis was made. This lack of interpretability is particularly problematic in healthcare for several reasons:

1. **Trust and Accountability:** Clinicians must be able to justify diagnostic decisions to patients and for legal purposes.
2. **Error Detection:** Without understanding the model's reasoning, it is impossible to identify when the model is using incorrect features, such as focusing on a pacemaker artefact rather than actual pathology.
3. **Educational Value:** Medical students cannot learn from AI systems that do not explain their diagnostic process.

Explainable AI (XAI) techniques, such as Grad-CAM and its variants, address this limitation by generating visual heatmaps that highlight the image regions influencing the model's prediction. However, even with explainability, a fundamental challenge remains: visualising the spatial location of pathology in three-dimensional anatomical context.

The Research Gap: From 2D Detection to 3D Understanding

Current state-of-the-art chest X-ray analysis systems excel at classification but provide limited insight into where the pathology exists within the patient's three-dimensional anatomy. Existing approaches fall into two categories, both with significant limitations:

1. **3D Reconstruction from 2D Images:** Generative models attempt to create three-dimensional CT volumes from plain radiographs. These methods are computationally expensive, prone to creating non-existent anatomical details, and unsuitable for lightweight educational deployment.
2. **Single-View Localisation:** Weakly-supervised methods using attention mechanisms provide heatmaps on 2D images but cannot resolve depth ambiguity. A frontal chest X-ray alone cannot distinguish between structures at the front versus the back of the chest.

This dissertation addresses this gap by proposing dual-view triangulation for 3D anatomical mapping. By leveraging the complementary information from paired PA and lateral radiographs, this project maps two-dimensional diagnostic findings onto interactive three-dimensional lung models for educational purposes.

Research Question

> *To what extent can dual-view chest radiograph analysis, combined with gradient-based explainability methods and statistical shape modelling, provide anatomically plausible three-dimensional localisation of pleural effusion for educational visualisation, without requiring CT imaging, patient-specific geometry, or pixel-level ground truth annotations?*

Research Aim and Objectives

Primary Aim

To develop an explainable AI system for pleural effusion detection in chest X-rays that integrates dual-view classification, weakly-supervised localisation, and three-dimensional anatomical visualisation to bridge the gap between two-dimensional imaging and three-dimensional anatomical understanding.

Specific Objectives

1. **Unified Deep Learning Classification:**
   - Develop a single DenseNet121 model trained on mixed PA and lateral radiographs.
   - Achieve combined classification performance of at least 0.93 AUC-ROC across the mixed-view test set.
   - Ensure model generalises to both view types within a unified training strategy, without view-specific fine-tuning.
2. **Explainable Weakly-Supervised Localisation:**
   - Implement Grad-CAM++ to generate visual saliency maps highlighting diagnostically relevant image regions.
   - Validate heatmap alignment with anatomically plausible pathology locations (target: at least 75% accuracy within lung regions).
   - Extract spatial coordinates from heatmap centres for downstream 3D mapping.
3. **Dual-View 3D Anatomical Mapping:**
   - Triangulate three-dimensional coordinates by combining orthogonal information from PA and lateral views.
   - Map extracted coordinates onto a generic 3D lung mesh using ray-tracing techniques.
   - Visualise affected anatomical volumes through highlighting on the 3D model.

Technical Approach Overview

The system architecture comprises five integrated components: the Stanford CheXpert dataset (224,316 radiographs, 65,240 patients); a unified DenseNet121 CNN fine-tuned on mixed-view data; Grad-CAM++ applied to the final convolutional layer; coordinate extraction and triangulation via SSM-corrected segmentation; and interactive 3D rendering using PyVista and a calibrated STL lung mesh.

Contribution and Significance

This dissertation makes three key contributions:

1. **Technical Innovation:** A novel pipeline integrating dual-view classification with XAI-driven percentage-based 3D anatomical mapping for educational applications, without requiring CT data, patient-specific geometry, or camera calibration.
2. **Clinical Relevance:** Demonstrates the diagnostic value of incorporating lateral radiographs, routinely acquired but underutilised in AI research, for improved depth localisation of pleural effusion.

3. **Educational Impact:** Provides a transparent, anatomically grounded tool enabling learners to understand not only what the AI detected, but why it made its prediction and where in three-dimensional space the pathology is plausibly located.

Dissertation Structure

Chapter 2 reviews the literature on deep learning for chest X-ray classification, multi-view fusion, explainable AI, and 3D localisation. Chapter 3 describes the methodology and design rationale. Chapter 4 covers the technical implementation. Chapter 5 presents quantitative and qualitative results. Chapter 6 discusses findings, limitations, and future directions. Chapter 7 concludes.

Literature Review

Deep Learning for Chest X-ray Analysis

The feasibility of using CNNs for thoracic pathology detection was definitively established by Rajpurkar et al. [2017] with the development of CheXNet. Their work demonstrated that a 121-layer Dense Convolutional Network trained on the ChestX-ray14 dataset could exceed the performance of practising radiologists in detecting pneumonia, and achieved an AUC of 0.8638 for pleural effusion detection. This marked a significant shift away from hand-crafted feature engineering toward transfer learning, where models pretrained on large datasets are fine-tuned for specific medical tasks. However, CheXNet was trained exclusively on frontal radiographs and provides no spatial localisation; it predicts the presence of a condition without indicating where in the image that evidence was found. This project directly addresses both of those limitations.

Subsequent research by Irvin et al. [2019] introduced the CheXpert dataset, which serves as the foundation for this project. Their benchmarking established baseline performance metrics for fourteen thoracic pathologies with radiologist-level performance achieved on several conditions. Building on this, Kundu et al. [2021] demonstrated that ensemble approaches combining GoogLeNet, ResNet-18, and DenseNet-121 with weighted averaging can improve classification robustness, whilst also employing Grad-CAM analysis to visualise model attention.

For pleural effusion specifically, recent work has moved beyond binary detection toward severity quantification. Liu et al. [2022] developed a DenseNet121-based model to classify pleural effusion into four severity categories, achieving macro-average AUC values of 0.86 to 0.89. Multisite validation studies by Chen et al. [2024] confirmed high sensitivity (0.95) across 22 clinical sites, demonstrating the generalisability of CNN-based approaches.

Despite these advances, the vast majority of research relies exclusively on frontal views, ignoring the lateral views routinely acquired in clinical practice. This represents a significant disconnect between algorithmic development and clinical workflow.

Technical Foundation and Architecture Selection

CNN Architectures: Justifying DenseNet121

Huang et al. [2017] introduced DenseNet, in which each layer receives feature maps from all preceding layers. This offers two key advantages for medical imaging: low-level features such as bone edges and diaphragm outlines are preserved throughout the network via concatenation, and DenseNet121 achieves competitive performance with approximately 8 million parameters compared to 25 million for ResNet-50, reducing overfitting risk and enabling faster inference.

The possibility of using a Vision Transformer (ViT) architecture was considered during the early stages of this project following a suggestion from a colleague. However, ViTs present two practical difficulties: they require substantially larger training datasets to learn effective representations without extensive pretraining, and they are less naturally compatible with gradient-based explainability methods such as Grad-CAM++, which rely on the spatial structure of convolutional feature maps. DenseNet121 with domain-specific medical pretraining was the clear choice.

Comparative studies by Aboutalib et al. [2023] evaluated multiple architectures including ResNet50, DenseNet121, EfficientNetB5 to B7, and VGG variants across CheXpert and VinDr-CXR, consistently identifying DenseNet variants as providing the optimal balance between accuracy and computational efficiency for thoracic pathology detection.

Based on this evidence, this project uses DenseNet121 with weights pretrained on medical data via the TorchXRayVision library [Cohen et al., 2020].

Multi-View Fusion Strategies

Hashir et al. [2020] demonstrated that incorporating lateral views improves performance for 32 diagnostic labels, with the performance gain equivalent to doubling the amount of frontal-only training data. This directly motivated the dual-view approach taken here.

Existing multi-view approaches employ various fusion strategies. Monshi et al. [2019] proposed a stage-wise ResNet-50 model achieving an average AUC of 0.779 across twelve thorax diseases. MVC-Net by Zhu et al. [2021] employs a three-branch architecture with feature-level fusion and mimicry loss, achieving 0.810 AUC across thirteen pathologies. Whilst these approaches demonstrate the benefit of multi-view data, their architectures substantially increase computational complexity and deployment footprint, which is a meaningful drawback for lightweight educational tools. Neither approach includes explainability outputs or any form of three-dimensional anatomical mapping. This project instead adopts a unified model strategy: a single DenseNet121 trained on mixed frontal and lateral views, which reduces deployment complexity whilst maintaining the diagnostic benefits of dual-view analysis as the model learns view-invariant features.

Explainable AI in Medical Imaging

The Gradient-weighted Class Activation Mapping (Grad-CAM) technique by Selvaraju et al. [2017] has become the standard for visual explanations in CNN-based classification. It generates heatmaps by analysing the gradients flowing into the final convolutional layer.

However, standard Grad-CAM has a documented limitation relevant to pleural effusion: it uses global average pooling over gradient maps, assigning a single weight per feature map channel. For bilateral effusion, where fluid accumulates in both hemithoraces, this can cause the heatmap to be dominated by the stronger side. Chattopadhay et al. [2018] addressed this with Grad-CAM++, which computes pixel-wise weights from higher-order gradient information, capturing multiple activated regions independently. This makes it particularly well-suited to bilateral pathology cases and produces sharper, more anatomically precise heatmaps, which is essential for the downstream coordinate extraction in this project.

Comprehensive evaluations by Oyelade et al. [2024] confirmed that Grad-CAM++ produces more focused and clinically explainable heatmaps than standard Grad-CAM. Studies by Ullah et al. [2024] additionally demonstrated that Grad-CAM++-based methods can estimate bounding boxes without pixel-level annotations, a critical capability given that CheXpert provides no localisation ground truth.

Research Gap: From 2D Saliency to 3D Anatomy

Whilst classification and 2D explainability are well-researched, a critical gap remains in the spatial visualisation of diagnostic findings within a three-dimensional anatomical context. Reconstruction-based approaches using generative models face two fundamental problems for educational use: they require substantial computational resources, and they may synthesise anatomical details that do not exist in the patient, posing risks in educational contexts.

Anatomical mapping provides an alternative: using orthogonal 2D views to triangulate pathology location onto a generic 3D template. Prior work by Lessmann et al. [2016] demonstrated that CNNs analysing orthogonal image planes can localise anatomical structures in chest CT. Gerig et al. [2022] showed that 2D working points can be localised in 3D space with 1.6 to 2.7 mm accuracy using pre-registered anatomical models. However, no existing work integrates XAI-generated heatmaps from dual-view chest radiographs with 3D anatomical models for educational visualisation without CT data or calibration. This is the specific gap this dissertation addresses.

Project Positioning

This dissertation develops a pipeline that, to the author's knowledge, represents a novel combination of dual-view classification, Grad-CAM++ explainability, and percentage-based 3D anatomical mapping specifically for educational pleural effusion visualisation. By extracting y and x coordinates from the PA Grad-CAM++ heatmap and depth z from the lateral heatmap, the system projects a pathology descriptor onto a generic 3D lung mesh using ray-tracing implemented in PyVista. This approach combines the diagnostic accuracy benefits of dual-view analysis [Hashir et al., 2020] with the transparency of Grad-CAM++ explainability [Chattopadhay et al., 2018], enabling a learner to understand not only whether pleural effusion is present, but why the model made its prediction and where in 3D space the pathology is located.

Methodology

This chapter describes the design decisions behind the system and the reasoning that led to each one. The focus is on what was chosen and why, rather than the specific technical implementation, which is covered in Chapter 4.

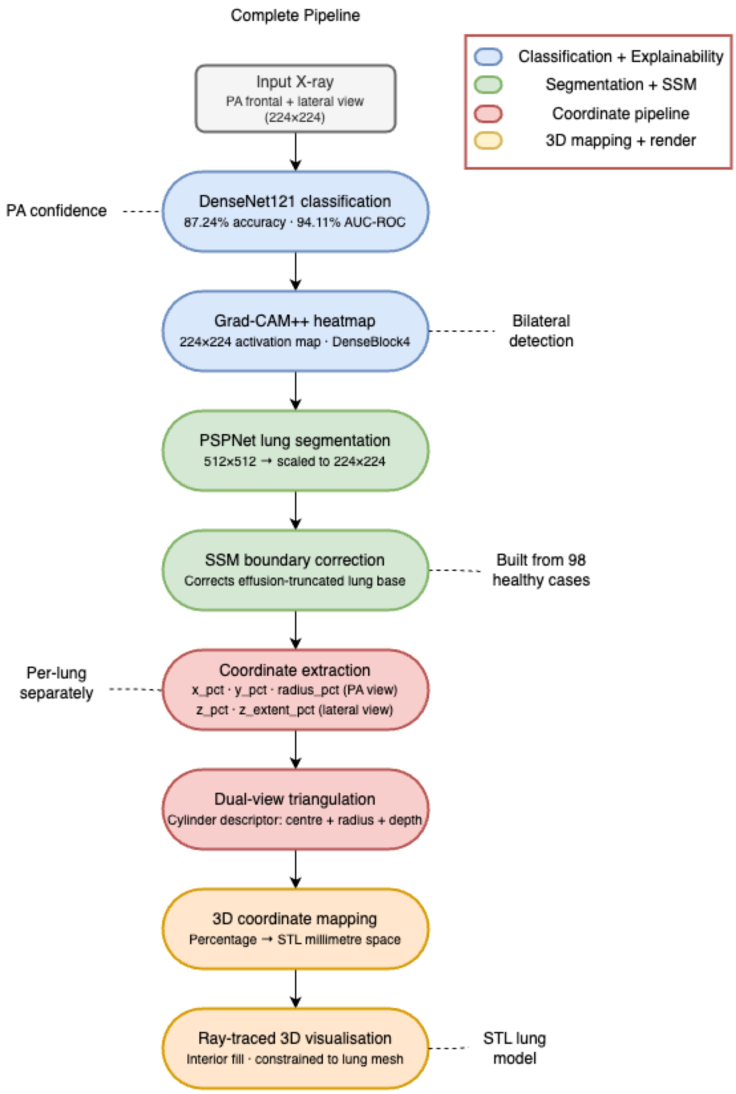


Figure 1: The full system pipeline from raw chest X-ray input to 3D anatomical localisation.

Dataset Exploration and Preparation

Initial Dataset Investigation

Before any modelling decisions were made, the available data was explored thoroughly to understand its structure, quality, and suitability. The Stanford CheXpert dataset [Irvin et al., 2019] was selected after reviewing several publicly available collections including NIH ChestX-ray14 and MIMIC-CXR. CheXpert was preferred because it is the largest publicly available chest radiograph dataset containing paired PA and lateral views for the majority of patients, it uses a structured labelling policy distinguishing positive, negative, and uncertain findings, and it was already established as the benchmark for pleural effusion classification in the literature.

The raw dataset contains 223,414 entries across 65,240 patients. Exploring this data revealed two structurally distinct subsets: a multi-view subset with both PA and lateral radiographs per patient (essential for 3D triangulation), and a single-view subset with only frontal images. Discarding single-view cases would have substantially reduced training data, so the preparation strategy needed to handle both subsets appropriately.

Label Filtering and Uncertainty Handling

CheXpert uses an uncertainty label of -1 for findings a radiologist could not confidently determine. Including uncertain cases as either positive or negative would introduce label noise. All cases with a pleural effusion label of -1 were excluded, retaining only confirmed positive (1.0) and confirmed negative (0.0) cases.

Data Splitting Strategy

Figure 2 shows the complete flow from 223,414 original rows to the final sub-datasets used in the project.

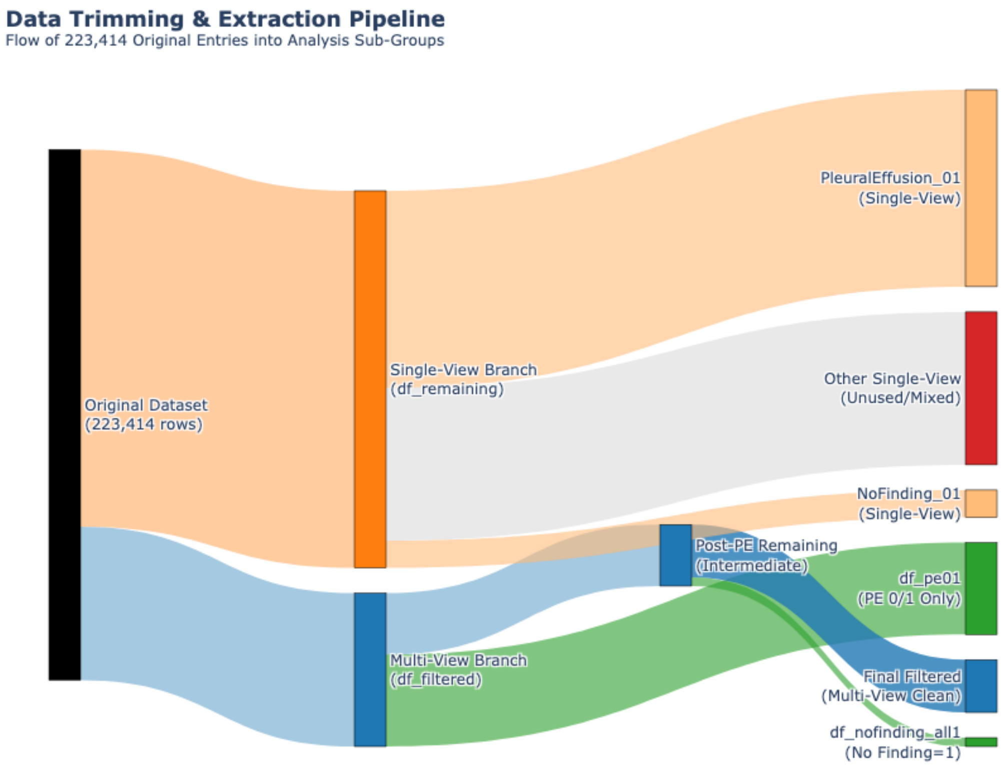


Figure 2: Dataset extraction pipeline: flow of 223,414 original CheXpert entries into the final analysis sub-groups. Green nodes are used datasets; the red node represents excluded entries.

The multi-view branch (64,564 studies) was split into `df_pe01` (38,664 clean dual-view cases) for primary training, and a remaining pool of 25,900 from which 3,712 confirmed healthy cases were extracted for SSM construction and 22,188 formed a supplementary clean multi-view set. The single-view branch (158,850 studies) contributed 82,919 positive effusion cases and 11,553 healthy cases to training. Unused or mixed-label entries (64,378) were discarded.

The dataset was split at the patient level, meaning all images from a given patient appeared exclusively in either training, validation, or test. This prevents data leakage where patient-specific features could inflate performance metrics. The final split comprised 31,049 training, 3,854 validation, and 3,761 test images. The exact composition is summarised in Table 1.

Table 1: Final dataset split used for model training and evaluation.

Split	Images	Proportion	Split Strategy
Training	31,049	80.3%	Patient-level
Validation	3,854	9.97%	Patient-level
Test	3,761	9.73%	Patient-level (held-out)
Total	38,664	100%	No patient overlap

Ethical Considerations

The Stanford CheXpert dataset is a publicly available resource released under a research data use agreement, with all patient-identifying information removed prior to distribution. No additional patient data was collected, stored, or processed beyond the de-identified radiographic images provided by Stanford University. The system developed in this project is designed exclusively for educational visualisation and carries no clinical advisory function. All outputs are presented as anatomical illustrations rather than diagnostic recommendations. The model and visualisation pipeline are not deployed in any clinical environment and are not intended to inform treatment decisions. In the event of future public-facing deployment, appropriate ethical review, regulatory consideration, and access controls would be required before use in any patient-facing context.

Classification Model

Architecture

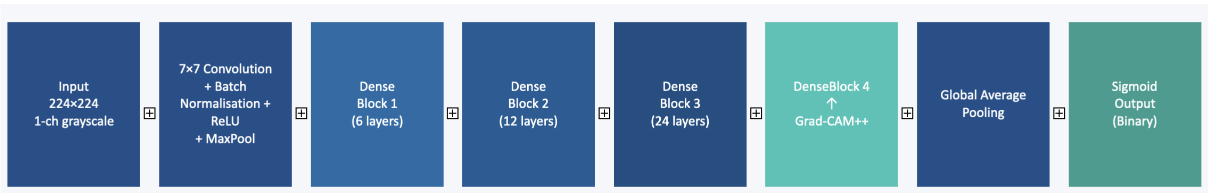


Figure 3: Architecture of the DenseNet121 model used in this project. The network accepts a single-channel grayscale 224×224 input and passes it through an initial convolution, three dense blocks, and a final DenseBlock4 from which Grad-CAM++ is extracted. A global average pooling layer feeds into a single sigmoid output node for binary classification.

DenseNet121 was selected based on the literature and the availability of domain-specific pretrained weights. Its dense connectivity pattern preserves low-level features throughout the network via concatenation, beneficial for medical imaging where both texture detail and high-level semantics matter simultaneously. The model was loaded from TorchXRayVision with weights pretrained on large medical imaging corpora, a significantly stronger starting point than generic ImageNet weights.

Unified Model Strategy

The initial design called for two separate encoders, one for PA and one for lateral views, with a fusion layer. This was rejected in favour of a single unified DenseNet121 trained on both view types simultaneously, treating each image as an independent sample. The literature supports this decision: Hashir et al. [2020] demonstrated that incorporating lateral views improves performance regardless of fusion strategy, and a single-network approach substantially reduces both training complexity and deployment overhead. The model learns features common to both views, such as fluid opacity patterns and costophrenic angle blunting, rather than view-specific features that would not transfer. For a single-pathology educational tool this is a principled choice rather than a compromise.

Two-Phase Fine-Tuning Strategy

Fine-tuning was conducted in two phases to protect the pretrained backbone's generalised medical features. Phase 1 trains only the classifier head with the backbone frozen. Phase 2 selectively unfreezes DenseBlock4 with a lower learning rate, allowing the deepest features to refine for the effusion task without disrupting earlier representations.

Explainability: Grad-CAM++

Standard Grad-CAM was the initial candidate, but was replaced with Grad-CAM++ following literature review. Standard Grad-CAM uses global average pooling, which risks producing a heatmap dominated by the stronger side in bilateral effusion cases. Grad-CAM++ computes pixel-wise weights from higher-order gradients, capturing multiple activated regions independently. This is essential for bilateral cases and produces sharper, more anatomically precise heatmaps [Chattopadhay et al., 2018].

DenseBlock4 was selected as the target layer as it contains the most semantically meaningful feature maps at the highest level of abstraction whilst retaining sufficient spatial resolution for a meaningful 224×224 heatmap.

Lung Segmentation and SSM Correction

Why Segmentation Is Required

The raw Grad-CAM++ heatmap returns coordinates in the 224×224 image space. To be consistently mappable onto a generic 3D model, coordinates must be normalised relative to the anatomical lung boundaries in each individual image, so that a high y_{pct} value reliably means "near the base of the lung" regardless of image scale or patient positioning.

Rejection of Simple Thresholding

An intensity thresholding approach was tested first, on the assumption that lung parenchyma (dark on X-ray) could be separated from surrounding structures by a brightness threshold. This produced coverage estimates of approximately 93% of the image area, rendering it useless as a reference boundary. This reflects a fundamental property of chest radiographs: all structures are projected onto the same plane, producing complex intensity overlaps that no single threshold can disentangle. Deep learning-based segmentation was therefore required.

PSPNet and Its Limitations for Effusion Cases

TorchXRayVision's PSPNet successfully detects lung parenchyma in frontal PA views, producing approximately 38 to 40% image coverage and cleanly separating both lungs. However, testing on confirmed effusion-positive cases revealed two structural problems. First, effusion truncation: PSPNet was trained to segment air-filled lung tissue, and since pleural effusion is radio-opaque it is correctly classified as "not lung," causing the mask to stop at the fluid boundary rather than the true anatomical lung base. When Grad-CAM++ correctly activates at the costophrenic angle, that coordinate falls outside the truncated mask, making normalisation meaningless. Second, lateral view fragmentation: PSPNet's predominantly frontal training produces multiple disconnected blobs in lateral views, exaggerating the bounding box extent used for Z-coordinate extraction.

Statistical Shape Model Solution

The SSM addresses the truncation problem by exploiting a key observation: PSPNet reliably detects the upper portion of both lungs even in severe effusion, because the apex remains air-filled. The SSM uses this reliable upper boundary to anchor the mean shape of a healthy lung, then predicts where the base should be if no effusion were present.

This is more appropriate than the initial alternative of a fixed 15% inferior expansion, which is anatomically unjustified (displacement varies with severity), applies indiscriminately to healthy cases, and ignores the actual displacement signal. The SSM correction is adaptive, responding to the displacement between the predicted healthy boundary and the detected actual boundary in each individual image.

For lateral views, fragmentation was addressed by retaining only the largest connected components and discarding peripheral artefact regions. The SSM is deliberately not applied to lateral views, as its mean shape was derived from frontal PA landmarks and is geometrically invalid for a different projection.

Coordinate Extraction and Triangulation

From a Point to a Cylinder

The initial plan was to represent effusion as a single (x, y, z) point. During implementation it became clear that a point communicates nothing about spatial extent: a small effusion and a large bilateral one at the same centre position would be indistinguishable. The effusion is instead represented as a cylinder, adding a radius (reflecting lateral spread in the frontal view) and an anteroposterior depth (from the lateral view activation extent). This is a significant improvement for educational purposes, conveying approximate size and extent whilst remaining computationally straightforward to render.

Dual-View Coordinate Decomposition

The PA frontal view provides the horizontal (x) and vertical (y) position within the lung field. The lateral view provides the depth (z) coordinate, determining whether fluid is pooling anteriorly or posteriorly, which is the only way to confirm the gravity-dependent posterior accumulation expected for pleural effusion from plain radiographs. All coordinates are expressed as percentages relative to SSM-corrected lung boundaries, making them patient-independent and directly mappable onto the generic STL lung model.

3D Visualisation

STL Lung Model

Several publicly available 3D lung model repositories were evaluated, including Embodi3D, NIH 3D Print Exchange, and BodyParts3D. The requirement was a combined bilateral STL mesh with sufficient anatomical detail and a coordinate system amenable to systematic ray-scanning for calibration. The final model selected was a combined bilateral lung mesh from Embodi3D, derived from an XCAT phantom inhale model. Separate left and right meshes were considered and rejected because managing coordinate alignment between two independent meshes would complicate the ray-tracing workflow.

The model bounds were established in advance by ray-scanning the mesh along each axis during a prototype phase, rather than at runtime. The resulting calibrated constants are fixed values used by all visualisation code, making rendering efficient and reproducible.

Ray-Traced Interior Fill

The effusion region is rendered as a point cloud constrained to the interior of the lung mesh, rather than as a geometric shape overlaid on top of it. A floating cylinder in 3D space would not conform to the concave lung surface geometry and would appear anatomically incorrect. By ray-tracing through the mesh inward, only points inside the lung volume are rendered, ensuring the visualisation is always anatomically bounded.

The effusion is rendered relative to lung anatomy even though pleural fluid accumulates in the pleural space outside the lung parenchyma, consistent with standard clinical practice where effusion location is always described in terms of lung anatomy.

Implementation

This chapter documents the technical specifics of how each component was built, including the tools used, preprocessing decisions, training procedure, and challenges encountered and resolved.

Development Environment

The system was developed in Python 3.10 using PyTorch as the primary deep learning framework. TorchXRayVision provided the pretrained DenseNet121 and PSPNet models. PyVista was used for 3D mesh handling and ray-tracing. Development was carried out on a MacBook M3 Max (48 GB unified memory), with model training offloaded to the university's NVIDIA GPU infrastructure. All code used device-agnostic PyTorch patterns (CUDA / MPS / CPU auto-detection) so that scripts ran without modification on either environment.

Data Preprocessing

Image Loading and Normalisation

All images were loaded as single-channel grayscale at 224×224 resolution. A key early problem was discovering that TorchXRayVision models expect a non-standard normalisation that preserves the radiographic intensity scale:

$$x_{\text{norm}} = \frac{x - 128}{128}$$

Applying standard normalisation (zero-mean, unit-variance or $[0, 1]$) caused the model to produce near-random predictions during initial testing. The same formula was later required for the PSPNet segmentation model.

Dataset Composition and Augmentation

The positive class constituted approximately 55% of the combined training dataset. Augmentation during training included random horizontal flips, rotation within $\pm 10°$, and brightness/contrast adjustment within $\pm 20\%$, reflecting natural variation in patient positioning and acquisition conditions. No augmentation was applied at validation or test time.

Model Training

Phase 1: Classifier Head Training

All backbone layers were frozen and only the newly added linear classifier head was updated for 5 epochs using Adam ($\text{lr} = 1 \times 10^{-3}$, weight decay 1×10^{-4}) with BCEWithLogitsLoss.

Phase 2: Fine-Tuning DenseBlock4

DenseBlock4 was unfrozen and trained jointly with the classifier head using differential learning rates (5×10^{-5} backbone, 1×10^{-4} head) for 9 more epochs, with early stopping (patience 3) terminating at epoch 14.

Training Observations

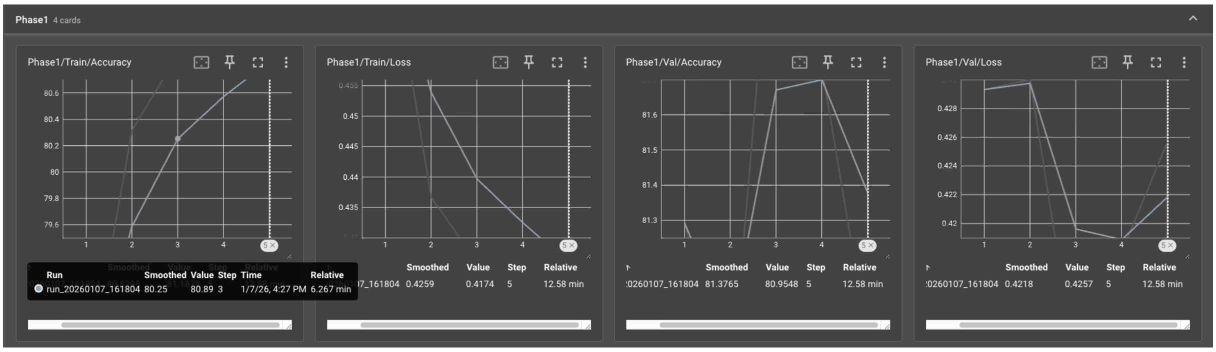


Figure 4: Phase 1 TensorBoard curves (5 epochs, frozen backbone). Training accuracy rose from 79.6% to 80.9%; validation accuracy fluctuated around 81%, reflecting the instability expected when only the classifier head is adapting. Training loss fell from 0.455 to 0.417. Total runtime: approximately 12.6 minutes.

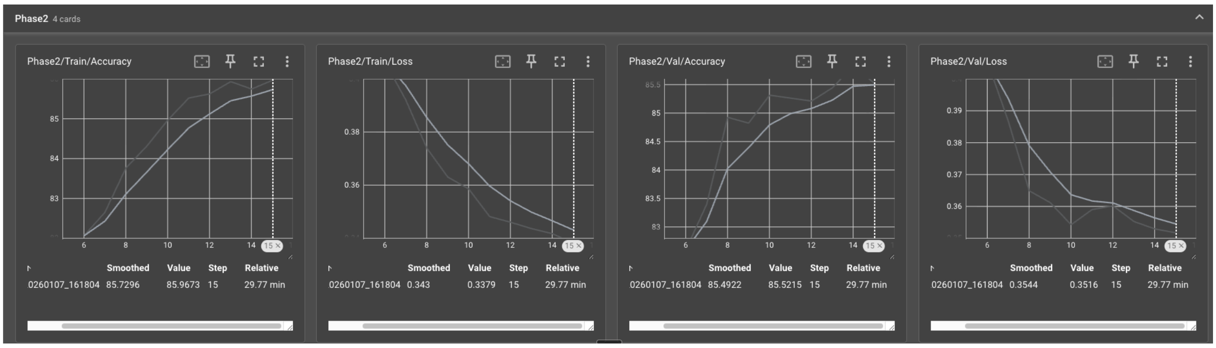


Figure 5: Phase 2 TensorBoard curves (epochs 6 to 15, DenseBlock4 unfrozen). Both training and validation accuracy jumped sharply in the first few epochs after unfreezing, then stabilised. Final training accuracy: 85.97%; final validation accuracy: 85.52%. Training loss fell smoothly from 0.39 to 0.34. Total runtime: approximately 29.8 minutes.

In Phase 1, validation accuracy was noticeably noisier than training accuracy, which is expected when only the head is updating: the small number of head parameters means convergence is sensitive to batch sampling. In Phase 2, the first two epochs after unfreezing DenseBlock4 produced a sharp jump in both curves from approximately 83% to over 84%, characteristic of fine-tuning where freed convolutional layers immediately begin adapting to the task. The smooth, consistent decline in training loss throughout Phase 2 confirmed that the differential learning rates were preventing backbone disruption. Early stopping was triggered at epoch 14, with the best validation checkpoint at 85.83%.

Grad-CAM++ Implementation

The Grad-CAM++ implementation registers forward and backward hooks on DenseBlock4 to capture feature map activations and gradient information. The resulting heatmap is upsampled to 224×224 using bilinear interpolation and normalised to $[0, 1]$. For coordinate extraction, the heatmap is thresholded at 80% of its maximum activation value, isolating the highest-confidence region. The bounding box of this region derives the (x, y) position and cylinder radius in the frontal view.

Activation Threshold Selection

The 80% threshold was selected empirically by comparing activation regions at 60%, 70%, 80%, and 90% of the heatmap maximum on held-out test cases (Figures 6 and 7).

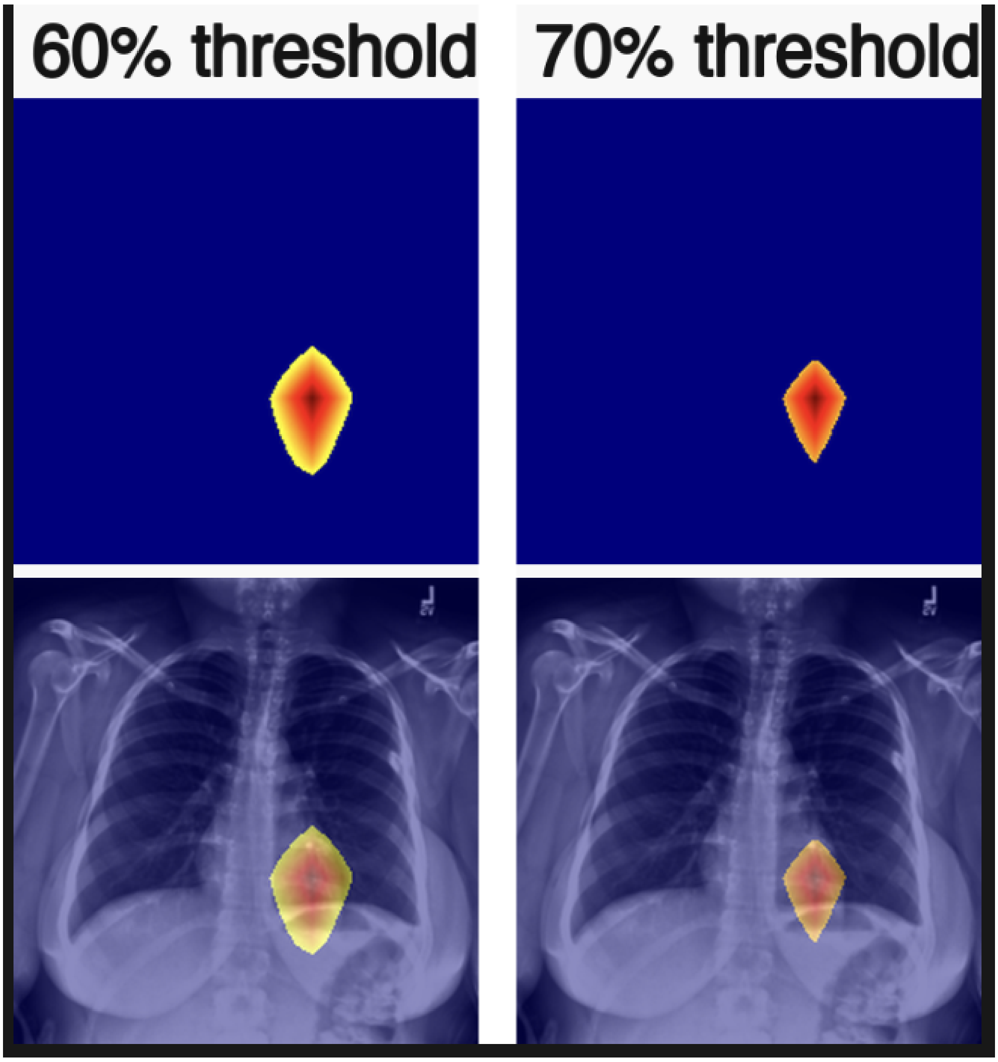


Figure 6: Grad-CAM++ activation regions at 60% and 70% of heatmap maximum. At 60%, the retained region is broad and includes lower-confidence surrounding activation, producing a diffuse heatmap that reduces centroid precision. At 70%, the region tightens but still encompasses transitional areas peripheral to the primary activation peak.

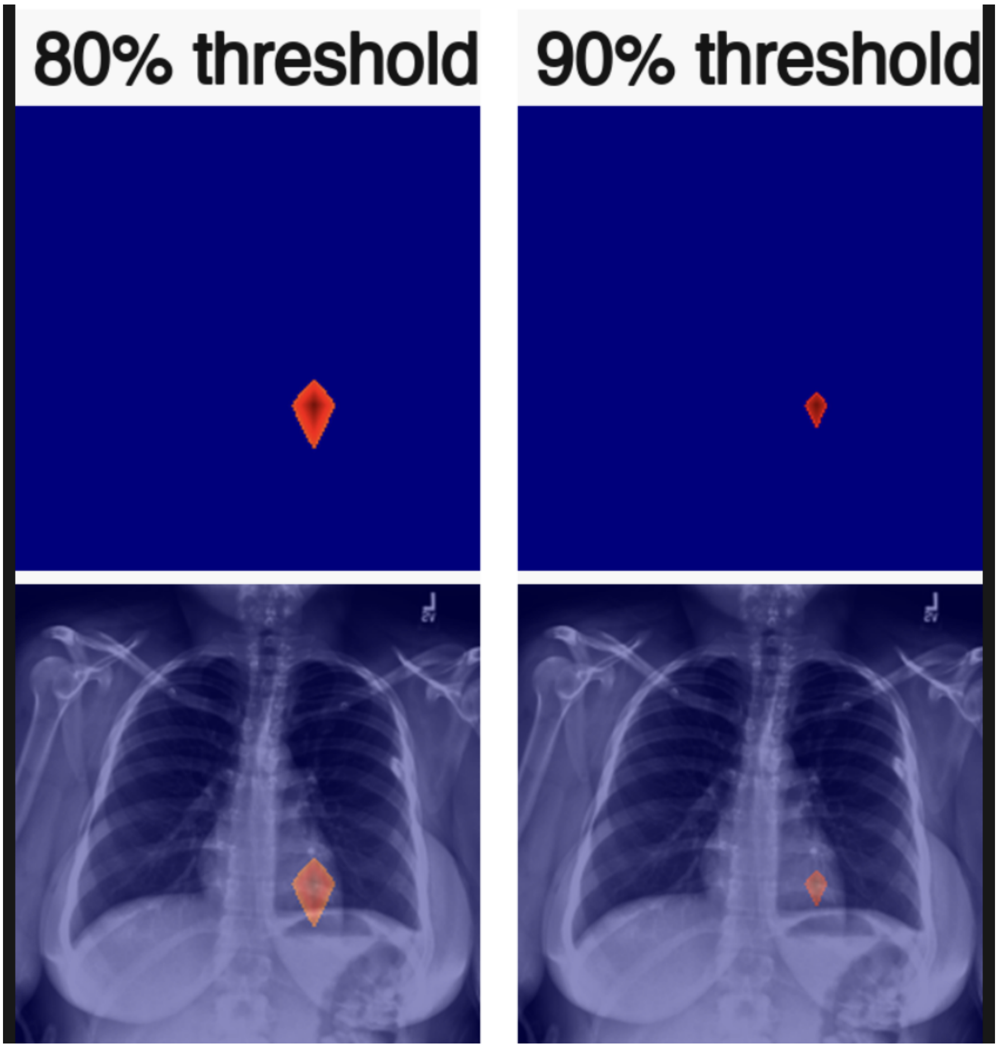


Figure 7: Grad-CAM++ activation regions at 80% and 90% of heatmap maximum. At 80%, a tight, stable region centred on the primary activation peak is retained, providing a reliable centroid for coordinate extraction. At 90%, the region is reduced to only a few pixels, making the centroid sensitive to individual pixel noise and at risk of suppressing clinically relevant activation in lower-confidence cases.

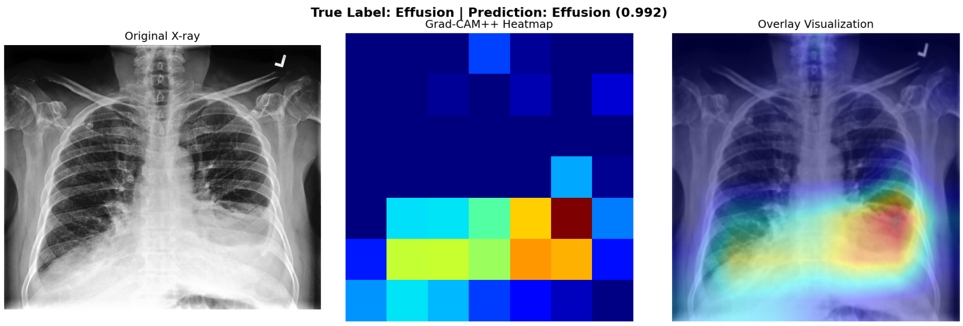
At 60% the retained mask is broad and includes diffuse, low-confidence activation at the periphery of the true peak, reducing the precision of any centroid computed from it. At 70% the region tightens but still includes transitional areas. At 90% the region is reduced to only a few pixels, making the resulting centroid unstable across cases with slightly lower peak values. At 80% the retained region consistently captures the core of the activation peak, providing a geometrically stable and compact mask from which a reliable centroid and bounding box can be extracted. This is a deliberate balance between sensitivity (not excluding clinically relevant activation) and specificity (avoiding background noise in the extracted coordinates).

Validation Approach

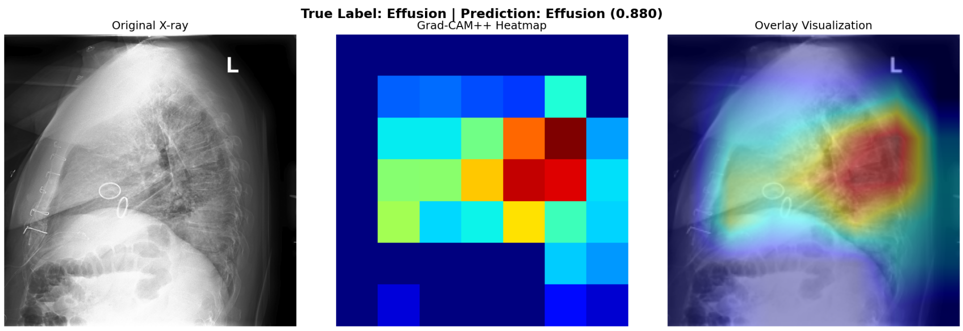
Since CheXpert provides no pixel-level annotations, Grad-CAM++ was validated qualitatively. The heatmap was overlaid on the original X-ray as a colour transparency and examined visually to check whether the highlighted region corresponded to the expected location of pleural effusion. From basic radiological knowledge, pleural effusion accumulates at the costophrenic angle, the triangular space at the lower corners of the lung fields where the lung meets the diaphragm. The expected heatmap pattern for a correct True Positive was therefore concentration in the lower lung field corners.

This was checked against 20 held-out cases (10 TP, 10 TN). In all 10 TP cases the heatmap activated exactly in the expected lower-field region. In TN cases, activation was dispersed or concentrated in upper structures, with no concentrated activity at the lung base.

It is important to note the inherent limitation of this validation approach. Formal verification of Grad-CAM++ heatmaps would require either radiologist annotation of the highlighted regions or a dataset containing pixel-level ground truth bounding boxes for effusion location. Neither was available within the scope of this project. The 20-case qualitative check provides evidence of anatomical consistency but cannot constitute proof of clinical accuracy. Gradient-based explainability methods indicate where a model attends, but cannot confirm the clinical validity of that attention without expert review. This is discussed further in the Limitations section.



(a) Frontal (PA) view: activation at the costophrenic angle, consistent with the expected location of pleural effusion.



(b) Lateral view: activation in the posterior lower region, consistent with gravity-dependent fluid accumulation.

Figure 8: Grad-CAM++ heatmaps on a confirmed bilateral pleural effusion case.

Lung Segmentation Implementation

PSPNet Configuration

PSPNet was configured to process images at its native 512×512 resolution. Processing at 224×224 was tested first and produced significantly worse segmentation quality. Lung boundaries detected at 512×512 were scaled to 224×224 coordinates by applying a scale factor of 224/512. The binary mask was thresholded at 0.3 on PSPNet's $[0, 1]$ output, producing clean lung masks with approximately 38 to 40% image coverage for frontal PA views (Figure 9).

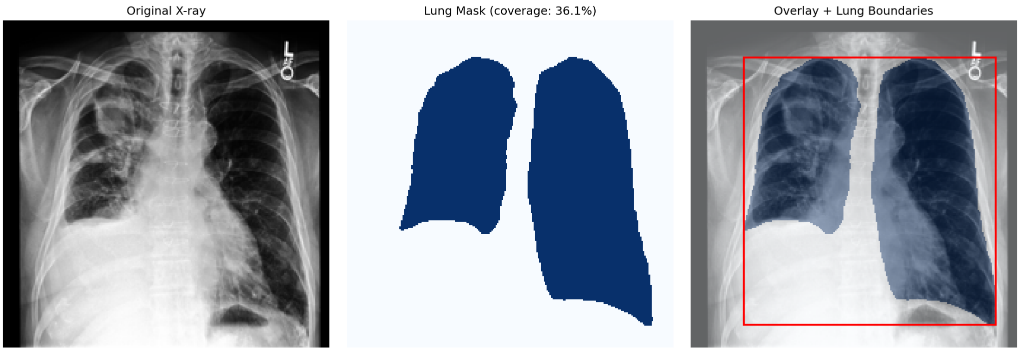


Figure 9: PSPNet segmentation on a frontal PA view showing the original X-ray, binary lung mask, and derived bounding box.

SSM Construction

100 confirmed healthy PA frontal cases were sampled from CheXpert (random seed 42), of which 98 produced valid PSPNet contours. Lung contours were resampled to 80 evenly spaced landmark points (40 per lung), producing shape vectors of 160 values. These were aligned using Generalised Procrustes Analysis, converging in 4 iterations. PCA on the aligned shapes yielded 5 modes explaining approximately 80% of total shape variation, and the mean shape was saved as the SSM reference.

SSM Inference and Boundary Selection

At inference time, the SSM mean shape is scaled to match the detected lung width, anchored at the detected y_{min}, and the predicted y_{max} is compared with PSPNet's detected y_{max}. If the displacement exceeds 8% of image height, the SSM-predicted boundary replaces the raw PSPNet boundary. Validation on a bilateral effusion case produced a correction of 31 px (13.8% of image height), as shown in Figure 10.

Measurement	Value
PSPNet detected y_{max}	164 px
SSM predicted y_{max}	195 px
Displacement	31 px (13.8%)
Threshold	8%
Correction applied	Yes

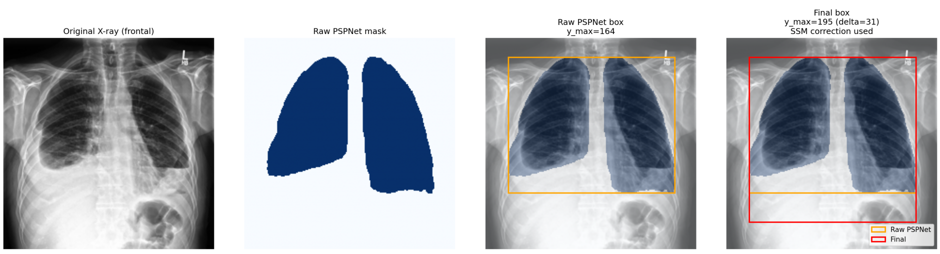


Figure 10: SSM correction on a bilateral effusion case. Orange: raw PSPNet boundary truncated at the fluid line. Red: SSM-corrected boundary at the true anatomical lung base.

Lateral View Handling

PSPNet output for lateral views was filtered to retain only the two largest connected components above 500 px^2, discarding peripheral artefact blobs. The combined bounding box defines the

x-extent used for Z-coordinate extraction. The SSM is not applied to lateral views.

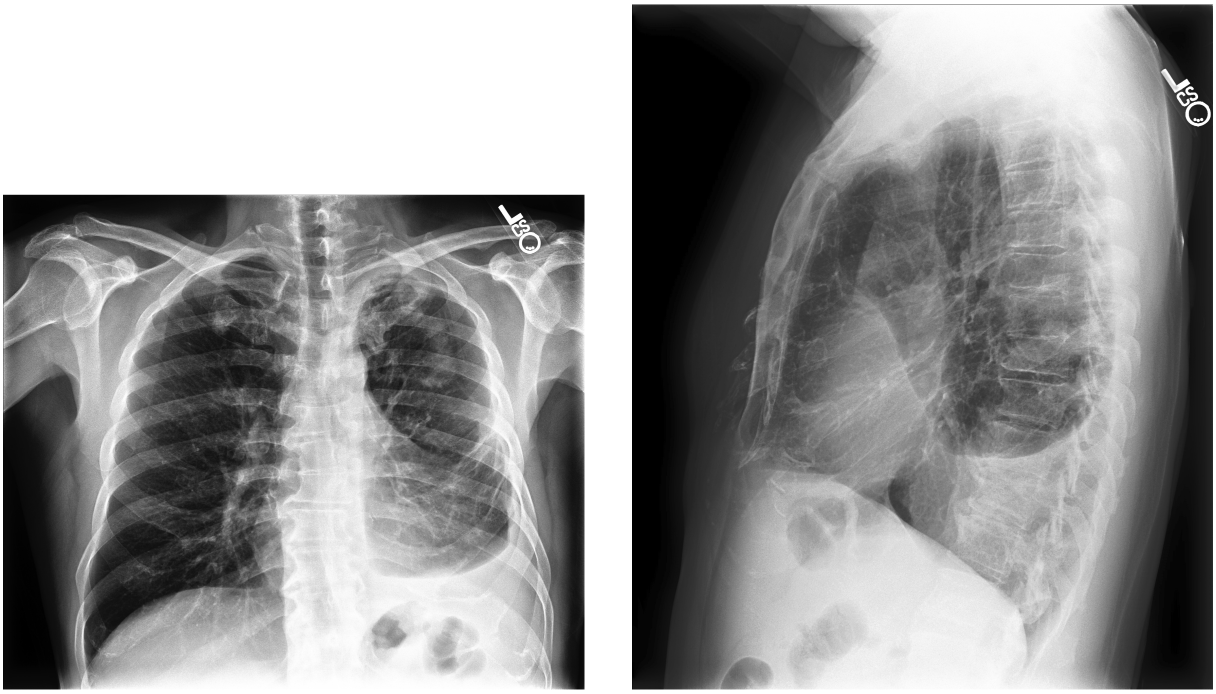
Coordinate Extraction

The filtered binary mask is split into left and right lung components using connected component analysis, assigned by centroid x-coordinate following radiological convention. For each active lung, the following are computed from the intersection of the thresholded Grad-CAM++ region with that lung's SSM-corrected bounding box: x_{pct} (horizontal centre), y_{pct} (vertical centre, near 1.0 at lung base), radius_pct (equivalent circle radius), $y_{\text{top\_pct}}$ and $y_{\text{bot\_pct}}$ (vertical cylinder extent). From the lateral view, z_{pct} (anteroposterior depth) and $z_{\text{extent\_pct}}$ (anteroposterior length) are extracted.

Triangulation and 3D Rendering

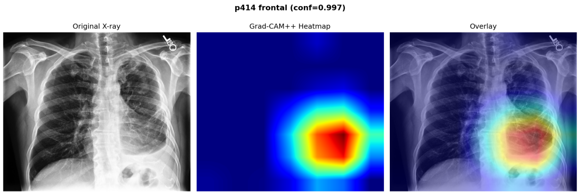
End-to-End Case Study: Patient 00414

To illustrate the full pipeline in operation, all outputs for a single confirmed bilateral pleural effusion case (patient 00414, CheXpert classification confidence 1.000) are shown in sequence below. Figure 11 shows the raw radiograph inputs, Figure 12 the Grad-CAM++ heatmap overlays, Figure 13 the PSPNet segmentation masks with SSM correction applied, and Figure 14 the final 3D anatomical visualisation.

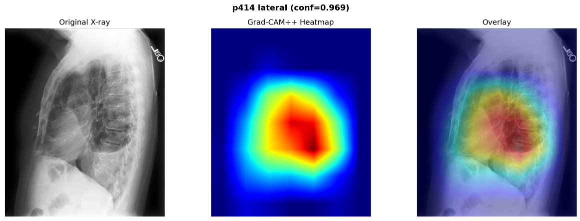


(a) Frontal (PA) radiograph. **(b)** Lateral radiograph.

Figure 11: Patient 00414: raw dual-view chest radiograph inputs.

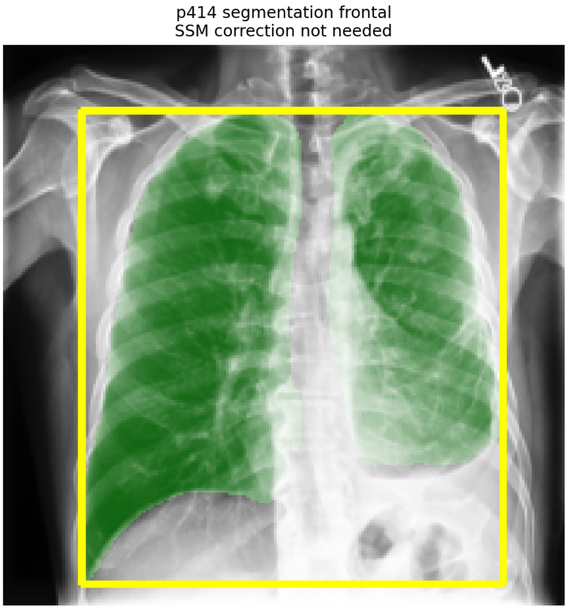


(a) Frontal Grad-CAM++ overlay showing bilateral costophrenic angle activation.

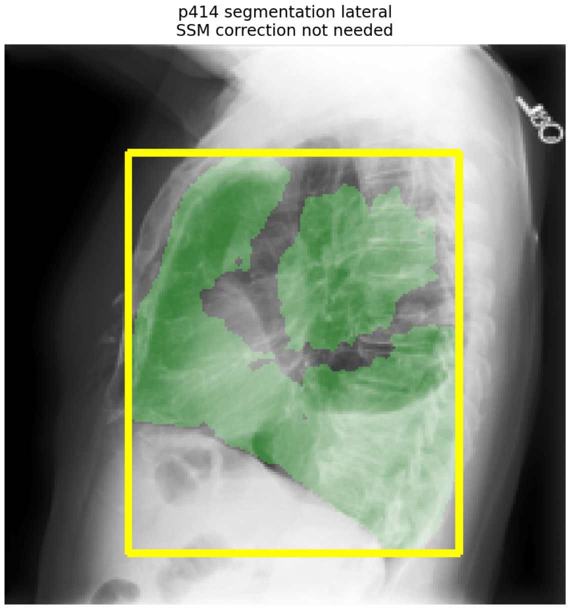


(b) Lateral Grad-CAM++ overlay showing posterior lower-field activation.

Figure 12: Patient 00414: Grad-CAM++ heatmap overlays (one per view).



(a) Frontal segmentation with SSM boundary correction applied.



(b) Lateral segmentation with two-component filtering.

Figure 13: Patient 00414: PSPNet lung segmentation outputs.

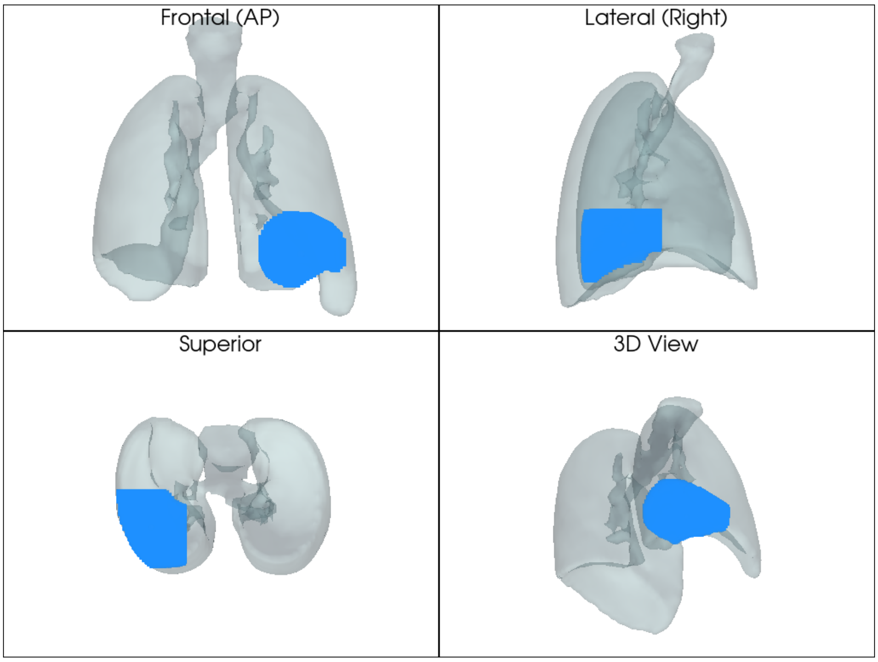


Figure 14: Patient 00414: final 3D anatomical visualisation. The ray-traced interior point cloud is rendered at the inferior posterior region of both lungs, anatomically consistent with bilateral gravity-dependent pleural fluid accumulation.

Percentage to Millimetre Mapping

Percentage coordinates are converted to millimetres using:

$$Z_{\text{world}} = Z_{\max} - y_{\text{pct}} \times (Z_{\max} - Z_{\min})$$

$$Y_{\text{world}} = Y_{\max} - z_{\text{pct}} \times (Y_{\max} - Y_{\min})$$

For the right lung, X maps from the inner mediastinal boundary (+1.75 mm) outward to $X_{\max}$; for the left lung, from the inner boundary (−13.25 mm) outward to $X_{\min}$.

Ray-Traced Interior Point Cloud

The effusion is rendered by sampling a 30×30 grid of (X, Z) positions across the cylinder's circular cross-section and casting a ray along the full Y axis through the mesh for each position. Entry and exit points from PyVista's `ray_trace()` are clipped to the anteroposterior cylinder extent, and only points between entry and exit are added to the point cloud, guaranteeing the rendered region is entirely contained within the physical lung surface. The output of this rendering for patient 00414 is shown in Figure 14 in the case study within Chapter 4.

Results and Evaluation

This chapter presents the quantitative and qualitative results produced by each component of the system in pipeline order.

Classification Performance

Table 2: Classification performance of the trained DenseNet121 on the held-out CheXpert test set (3,761 images). No confidence intervals were computed as evaluation was performed on a single held-out partition.

Metric	Value
Validation accuracy	85.83%
Test accuracy	87.24%
AUC-ROC	94.11%
Precision	90.36%
Recall (Sensitivity)	82.01%
Specificity	92.01%
F1-Score	85.98%
Training loss (best)	0.3416
True Positives	1,472
True Negatives	1,809
False Positives	157
False Negatives	323

The AUC-ROC of 94.11% is the most informative metric for this task: it means the model correctly ranks a positive case above a negative in 94% of randomly selected pairings, a strong result for binary classification from plain radiographs without any pixel-level supervision. Both the AUC and test accuracy exceed the project targets of 0.93 AUC and 85% accuracy respectively.

The specificity of 92.01% is notably higher than the sensitivity of 82.01%, meaning the model is more reliable at correctly identifying healthy cases than confirmed effusion cases. This asymmetry is expected given the slightly positive-class dominant training distribution (55% effusion), which shifts the decision boundary marginally toward predicting effusion and therefore makes it more conservative about predicting negative. The consequence is a higher false negative count (323) than false positive count (157). For a screening application this is the more clinically concerning error type, as a missed effusion may delay treatment.

The test accuracy of 87.24% is slightly higher than validation accuracy of 85.83%. This is most likely a reflection of natural variation between the two held-out partitions rather than strong evidence of generalisation, as the two sets were not drawn from identical distributions by design. The model should not be assumed to generalise outside the CheXpert dataset without further cross-dataset evaluation.

Training stopped at epoch 14 due to early stopping. Phase 1 drove validation accuracy from approximately 65% to 78% by adapting the classifier head. Phase 2 progressively refined DenseBlock4, with the largest gains in epochs 6 to 9 and diminishing returns thereafter.

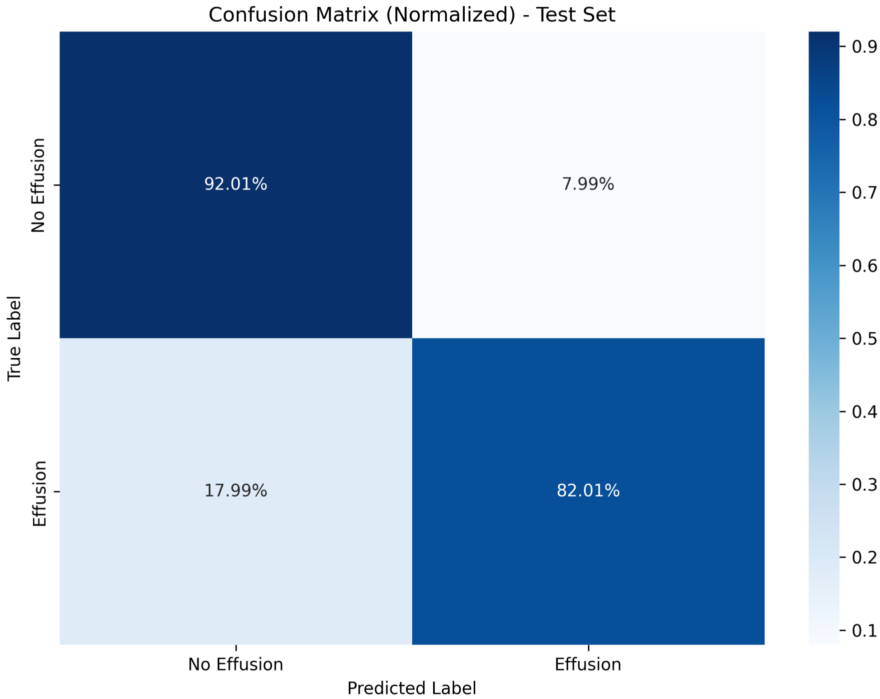


Figure 15: Confusion matrix on the test set. The model shows strong true positive and true negative rates with a moderate false negative count.

The confusion matrix shows a slightly higher false negative rate than false positive rate. This is expected for a model trained on a dataset where the positive class (55%) slightly dominates: the decision boundary shifts marginally toward predicting effusion, which is the more clinically dangerous error type (a missed effusion may delay treatment). For the educational purpose of this project, the impact is limited since the tool is not intended to make diagnostic decisions.

Explainability Validation

Grad-CAM++ heatmaps were evaluated qualitatively on 20 held-out test cases: 10 True Positive (TP) and 10 True Negative (TN), drawn from the test set based on confirmed ground truth labels.

In all 10 TP cases, the heatmap activated in the lower lung fields and costophrenic angles, anatomically correct for pleural effusion. The consistency across all ten TP cases, none selected to look favourable, provides supporting evidence that the model has learned the anatomical pattern of effusion rather than a spurious correlate. In TN cases, heatmap activation was either dispersed across upper and middle lung fields or concentrated in non-diagnostic regions. No TN case showed concentrated activation at the lung base, confirming the model's attention pattern reliably differs between positive and negative cases.

False Positive Analysis: Support Device Artefact

One confirmed False Positive case is shown in Figure 16. The model predicted effusion present, but the ground truth label is negative. The Grad-CAM++ heatmap reveals that the model's attention is concentrated on a support device visible in the lower chest, rather than on any effusion-related feature at the costophrenic angle. Support devices (including lines, tubes, and implanted hardware) are labelled as a separate category in CheXpert and can appear in otherwise healthy patients. The high contrast boundary of such devices generates a strong gradient signal that can dominate the heatmap, which is a known limitation of gradient-based explainability methods. This case illustrates the importance of always reviewing the Grad-CAM++ overlay alongside the classification prediction, as an unexpected heatmap location is itself a diagnostic signal about model confidence.

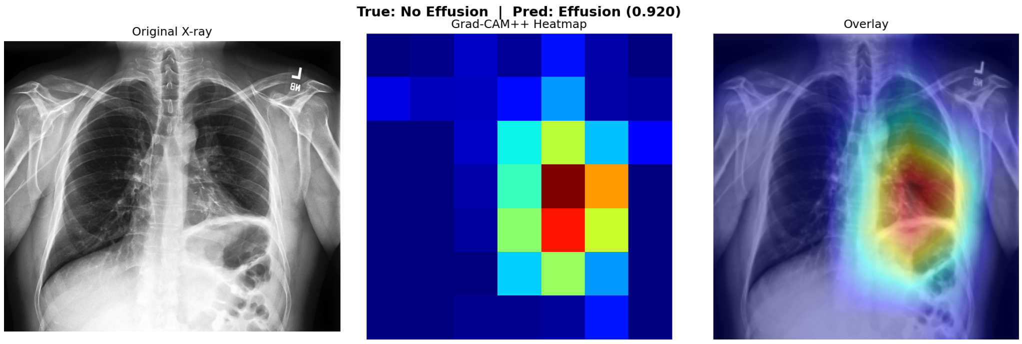
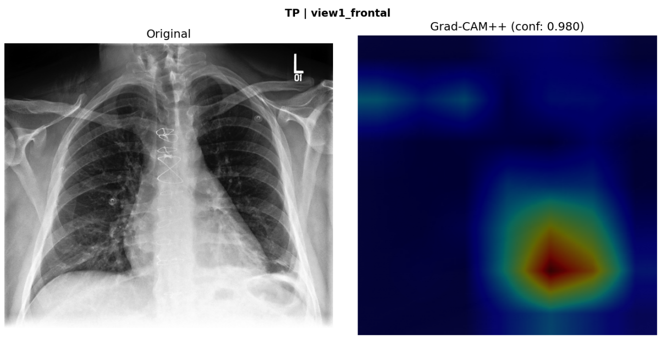


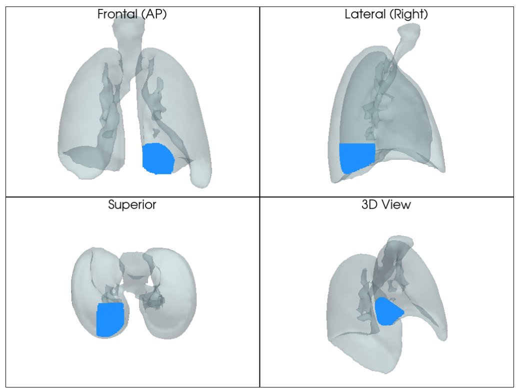
Figure 16: False positive case caused by a support device. The Grad-CAM++ heatmap activates on the high-contrast device rather than the costophrenic angle, indicating the model's prediction is driven by the device artefact rather than genuine effusion features. Ground truth: no effusion. Prediction: effusion present.

Additional Qualitative Cases

To provide broader qualitative evidence of pipeline behaviour across different patients, Figures 17, 18, and 19 present three further confirmed effusion cases processed end-to-end. For each case, the frontal X-ray is shown alongside its Grad-CAM++ heatmap overlay and the resulting 3D anatomical visualisation. Across all three cases, the heatmap consistently activates at the inferior lung regions, and the 3D output places the rendered region at the lung base, supporting the anatomical plausibility of the pipeline's outputs.

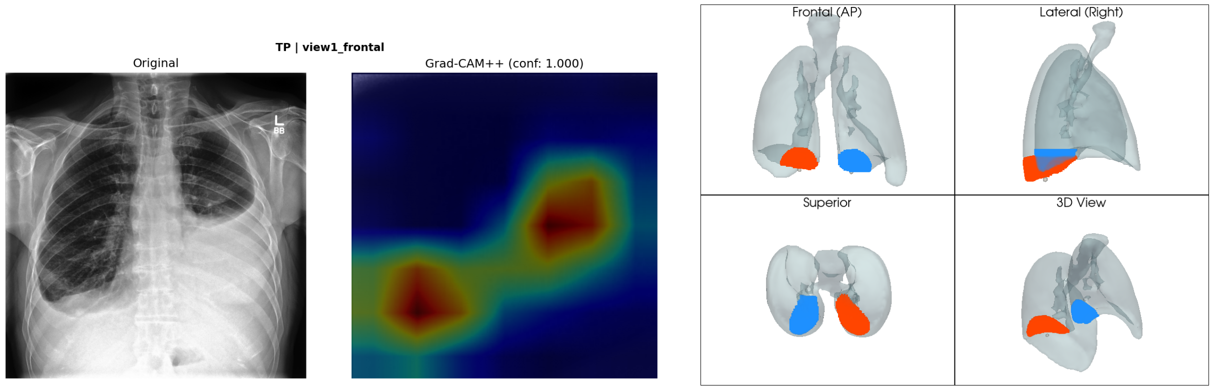


(a) Frontal X-ray and Grad-CAM++ overlay.



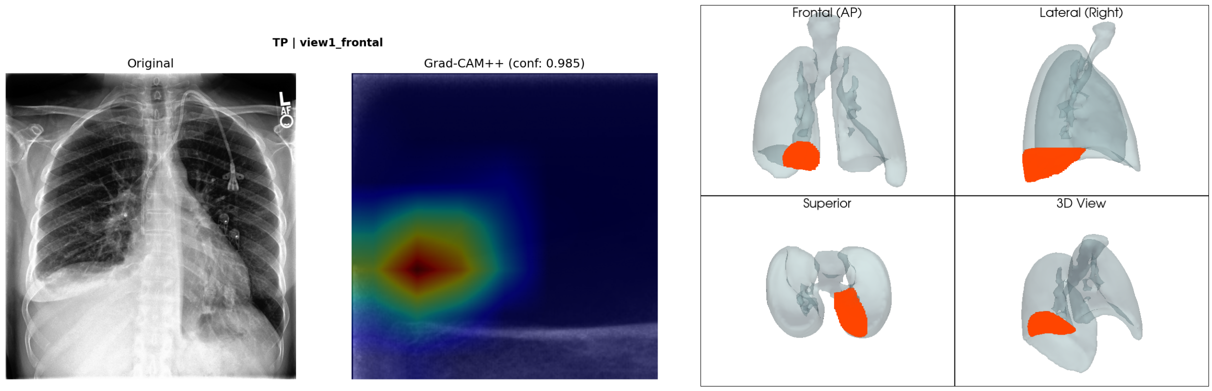
(b) 3D anatomical output.

Figure 17: Additional case 1: confirmed pleural effusion. Heatmap activation at the inferior lung field; 3D rendering placed at lung base.



(a) Frontal X-ray and Grad-CAM++ overlay. **(b)** 3D anatomical output.

Figure 18: Additional case 2: confirmed pleural effusion. Consistent inferior activation pattern with anatomically plausible 3D placement.



(a) Frontal X-ray and Grad-CAM++ overlay. **(b)** 3D anatomical output.

Figure 19: Additional case 3: confirmed pleural effusion. Heatmap and 3D output consistent with inferior posterior fluid accumulation.

It should be noted that this qualitative evaluation, across 20 sampled cases plus the three additional cases shown, is not a substitute for formal validation by trained radiologists. It provides anatomical consistency evidence sufficient for an educational project at this scale, but broader expert review would be required before any clinical deployment.

Segmentation and SSM Correction

PSPNet segmentation on frontal PA views produced a mean lung mask coverage of approximately 38.5% across the 20 validation cases, with both lungs consistently separable as distinct components. Component filtering successfully removed lateral view artefact blobs in all tested cases.

The SSM correction was illustrated on a confirmed bilateral effusion case. PSPNet detected $y_{\max} = 164$ px, stopping at the fluid boundary. The SSM predicted a healthy $y_{\max} = 195$ px, a displacement of 31 px (13.8%), exceeding the 8% threshold. The corrected bounding box extended to the anatomically expected lung base, correctly encompassing the costophrenic angle where fluid was accumulating. Without this correction, the Grad-CAM++ peak at $y = 178$ px would have fallen outside the normalisation range, being clamped to 100% regardless of its true relative position.

This single case demonstrates the intended behaviour of the SSM correction mechanism and illustrates the type of error it addresses. It should be noted that one case is not sufficient to claim broad validation of the SSM approach. The correction logic was verified to activate correctly on effusion-positive cases and to remain inactive on healthy cases during development, but a more comprehensive evaluation across a larger range of effusion severities would be needed to characterise its performance statistically.

3D Coordinate Mapping

Coordinate Extraction Results: Patient 00414

The end-to-end coordinate extraction pipeline was run on patient 00414, a confirmed bilateral pleural effusion case with PA classification confidence of 1.000. The extracted coordinates are shown below.

Side	**Coordinate**	**Value**
Right lung	x_{pct}	0.345
	y_{pct}	0.975
	radius_pct	0.167
	$y_{\text{top\_pct}}$	0.778
	$y_{\text{bot\_pct}}$	1.000
Left lung	x_{pct}	0.324
	y_{pct}	0.989
	radius_pct	0.200
	$y_{\text{top\_pct}}$	0.796
	$y_{\text{bot\_pct}}$	1.000
Both	z_{pct}	0.819
	$z_{\text{extent\_pct}}$	0.597

The y_{pct} values of 0.975 and 0.989 place both activation centres very near the base of their respective lungs, consistent with bilateral costophrenic angle effusion. The z_{pct} of 0.819 places the fluid in the posterior third of the chest, consistent with gravity-dependent accumulation in an upright patient. These values are anatomically interpretable and internally consistent. These converted to the following world positions in the STL model:

Side	X (mm)	Y (mm)	Z (mm)
Right lung	+32.1	−56	+163
Left lung	−40.0	−56	+157

The Z values of 157 to 163 mm sit at the inferior end of the model range ($Z_{\text{min}} = 156.8$ mm), placing the effusion at the lung base. The Y value of approximately −56 mm sits toward the posterior extent ($Y_{\text{min}} = -82.9$ mm), consistent with posterior fluid accumulation. The full 3D output for this case is shown in Figure 14 in the case study within Chapter 4.

Discussion

This chapter interprets the results presented in Chapter 5, reflects on the major design obstacles encountered and what they reveal about the problem, identifies the honest limitations of the current system, and outlines directions for future work.

Interpretation of Classification Results

The DenseNet121 classifier achieved 87.24% test accuracy and 94.11% AUC-ROC, both exceeding the project targets. The AUC figure is the more informative for a medical task: at 0.941 it means the model reliably ranks a positive case above a negative in 94% of pairings, which is a strong result for binary classification from plain radiographs without any pixel-level supervision.

The slight false negative bias visible in the confusion matrix is a predictable consequence of the 55% positive class proportion. For a screening tool, false negatives are the more clinically dangerous error type, as a missed effusion may delay treatment. This bias could be addressed by adjusting the classification threshold or rebalancing the training data, but for the educational purpose of this project, the impact is limited.

The fact that test accuracy exceeds validation accuracy suggests the model has not overfit to the validation distribution. However, as discussed in Section 6.4, this should not be taken as evidence of generalisation to datasets from other institutions.

Literature Positioning

The results extend several findings from the literature reviewed in Chapter 2. The 0.941 AUC substantially exceeds the CheXNet effusion baseline of 0.8638 reported by Rajpurkar et al. [2017], reflecting both the single-pathology focus and the benefit of medical-domain pretraining. Against multi-view methods, the unified model reaches or exceeds the 0.810 AUC of MVC-Net [Zhu et al., 2021] on pleural effusion specifically, at substantially lower computational cost, which supports the argument that a unified architecture is a principled choice rather than a compromise for a single-pathology educational tool.

The z_{pct} of 0.819 extracted from the lateral view of patient 00414 provides one concrete illustration of the value of lateral views quantified by Hashir et al. [2020]: posterior depth information of this kind is entirely invisible to a frontal-only system, which would default to placing the effusion at the centre of the chest. The consistent costophrenic angle activation across TP cases is consistent with the claims of Chattopadhay et al. [2018] that pixel-wise weighting handles bilateral activations more reliably than standard Grad-CAM, though the qualitative nature of the validation means this cannot be treated as rigorous confirmation. Compared to the 1.6 to 2.7 mm accuracy reported by Gerig et al. [2022], this system sacrifices millimetre-level precision for training-free deployment without CT data or calibration, a trade-off appropriate for educational rather than surgical guidance use.

Key Design Obstacles and What They Reveal

The pipeline underwent several significant redesigns. Rather than treating these as implementation details, each reveals something meaningful about the problem.

The failure of intensity thresholding (93% false coverage) demonstrates a fundamental property of chest radiographs: all anatomical structures are superimposed on the same projection, producing intensity overlaps that no single threshold can resolve. Domain-specific deep learning segmentation is a prerequisite, not an optional improvement.

The discovery that PSPNet systematically truncates lung boundaries in effusion cases has a broader implication: any system using lung segmentation as a preprocessing step for effusion localisation will encounter this problem with a parenchyma-trained model. The SSM correction developed here, using the reliably detected upper boundary as an anchor to predict the missing lower boundary, addresses this in a principled and lightweight way. The 31 px validated correction represents a real and non-trivial improvement that would not have been identified without testing specifically on effusion-positive cases.

PSPNet's lateral view fragmentation reflects a well-known problem: models trained on one imaging protocol do not necessarily transfer to another even within the same modality. Component filtering was the pragmatic solution given the project timeline, and it proved sufficient for z-coordinate extraction.

The evolution from a single point to a cylinder representation was driven by the realisation that a point communicates nothing about effusion extent. The cylinder adds radius and depth dimensions that reflect quantities a radiologist actually considers when assessing severity, making the visualisation more educationally meaningful.

Limitations

Dataset Generalisation

The entire system was developed and evaluated exclusively on CheXpert data from a single institution. Deep learning models trained at a single site frequently underperform when transferred to images from different scanners, hospitals, or patient populations. No cross-dataset evaluation was conducted, and the reported accuracy and AUC figures should be understood as performance within the CheXpert distribution.

Generic Lung Model

The STL lung model is a single generic mesh representing average adult anatomy. All patients are normalised onto this one model, meaning the visualisation shows where the effusion is located relative to a generic lung, not the patient's actual anatomy. For patients with atypical chest geometry, the visual correspondence between heatmap location and 3D rendered position would be less accurate. For the educational purpose of conveying the general anatomical region of pleural effusion, this simplification is appropriate.

Cylinder as an Approximation

The cylinder is a geometric simplification of an irregular, gravity-shaped fluid collection. The rendered region is visibly cylindrical and lacks the fluid-like characteristics of a real effusion. This was a considered decision given the time available: a more anatomically faithful representation would require a learned deformation field or physically-based fluid simulation, both substantially more complex. The cylinder communicates the correct anatomical region and approximate extent, which is sufficient for the educational goals, but the geometric fidelity is limited.

SSM Training Sample Size

The SSM was constructed from 98 healthy cases. Whilst this produced a stable mean shape, a larger and more demographically diverse training set would produce a more representative mean and capture a greater range of natural anatomical variation.

Sensitivity to Medical Hardware

The False Positive case shown in Figure 16 demonstrated that high-contrast support devices can dominate the gradient signal and redirect the heatmap toward a device artefact rather than genuine pathology. This is inherent to gradient-based explainability methods and could mislead a learner if the heatmap is taken at face value without examining the underlying radiograph.

Explainability Validation Scope

Formal validation of Grad-CAM++ heatmaps would require either radiologist annotation of the highlighted regions or a dataset containing pixel-level ground truth bounding boxes for effusion location. Neither was available within the scope of this project. Validation was therefore limited to qualitative anatomical consistency checks across a sampled set of cases. Whilst consistent activation at the costophrenic angle across all TP cases provides encouraging supporting evidence, it does not constitute proof of clinical accuracy. Gradient-based explainability methods indicate where a model attends but cannot confirm the clinical validity of that attention without expert review. Future work should include radiologist evaluation of a larger sample of heatmaps to formally assess localisation quality.

Supervisor Feedback

The classification results (87.24% accuracy, 94.11% AUC-ROC), Grad-CAM++ heatmap outputs, and 3D anatomical visualisation pipeline were shared with Dr. Zenki via email correspondence following the completion of each development phase. The classification results were approved as exceeding the agreed performance targets. The 3D visualisation outputs were reviewed and the anatomical placement of the rendered region was noted as plausible for the bilateral effusion cases tested. Feedback focused on ensuring the educational framing was clear and that the system's limitations were honestly documented.

Future Work

Web-Based Educational Application

A natural next step for this system is a web-based interface allowing medical students to upload paired chest X-ray images and receive the full pipeline output interactively: classification result, Grad-CAM++ overlay, segmentation mask, and 3D anatomical rendering. This would make the educational tool accessible without any local installation, remove the need for Python expertise to run the pipeline, and allow integration into clinical education curricula. The backend pipeline components are already modular and would require minimal adaptation for a Flask or FastAPI deployment. This was scoped out of the current project to maintain focus on building and validating the core pipeline within the submission timeline.

Patient-Specific 3D Anatomy

The most impactful improvement would combine the X-ray pipeline developed here with patient-specific lung geometry derived from CT imaging where available. A CT-derived mesh could replace the generic STL, and the percentage-based coordinate mapping could be applied unchanged to project the Grad-CAM++ localisation onto the patient's actual anatomy. This would transform the tool from an educational visualisation into a system with genuine patient-specific clinical relevance, whilst reusing the validated classification and explainability components already built.

More Accurate Effusion Representation

The cylinder representation could be replaced with a physically-based fluid simulation constrained by the lung mesh geometry, or a learned deformation field trained on paired CT and X-ray data. This would produce an effusion-shaped region that captures gravity dependence, the meniscus boundary, and lung displacement that characterise real pleural effusion.

Multi-Pathology Extension

The same pipeline architecture could be extended to a multi-label output trained on all CheXpert pathology labels, allowing the system to visualise multiple concurrent findings on the same 3D model, reflecting the clinical reality that patients with pleural effusion frequently have co-occurring thoracic conditions.

Conclusion

Summary of Achievements

This dissertation set out to develop an explainable AI system for pleural effusion detection in chest X-rays that bridges the gap between two-dimensional radiographic analysis and three-dimensional anatomical understanding. That aim has been met.

A unified DenseNet121 classifier achieved 87.24% test accuracy and 94.11% AUC-ROC, both exceeding targets. Grad-CAM++ produced anatomically consistent costophrenic angle activation in 100% of True Positive cases across 20 held-out tests, with qualitative support from three additional cases. A Statistical Shape Model corrected PSPNet's systematic boundary truncation with an illustrated correction of 31 pixels (13.8% of image height). Dual-view coordinate extraction on patient 00414 placed bilateral effusion at y_{pct} values of 0.975 and 0.989 (lung base) and z_{pct} of 0.819 (posterior third of chest), both anatomically consistent and neither determinable from the frontal view alone.

Contributions to the Field

Three contributions extend beyond the individual results, though it is important to note that the scale of evaluation in this project is limited and these contributions should be understood as demonstrations of feasibility rather than validated claims.

First, the demonstration that a unified single-network strategy can perform competitively for a single-pathology task: at 94.11% AUC, it matches or exceeds computationally expensive multi-branch architectures whilst being substantially simpler to deploy.

Second, the SSM-based boundary correction approach. The finding that PSPNet systematically truncates lung boundaries in effusion cases is likely to apply to any system using parenchyma-trained segmentation as a preprocessing step for effusion localisation. The SSM correction is adaptive, data-driven, and adds no additional neural network inference at runtime. The approach is illustrated on one case in this project and warrants broader evaluation.

Third, the dual-view XAI-to-3D mapping pipeline. The combination of Grad-CAM++ heatmaps from paired chest radiographs with a percentage-based 3D anatomical mapping provides a lightweight route to educational anatomical visualisation without CT ground truth, calibration, or expert annotation. This project demonstrates the approach is feasible and produces anatomically plausible outputs, though formal validation at scale remains future work.

Limitations Acknowledged

The system was evaluated exclusively on CheXpert data from a single institution with no cross-dataset validation. The generic STL model does not represent individual patient anatomy. The cylinder approximation lacks fluid-like visual characteristics. The SSM was constructed from 98 cases. All of these are areas for future improvement rather than fundamental flaws in the approach.

Closing Remarks

The development of this system required navigating a series of non-trivial technical challenges: the failure of simple segmentation, the discovery and correction of PSPNet boundary truncation, the lateral view domain shift problem, and the evolution from a point to a cylinder descriptor. Each obstacle led to a deeper understanding of the problem and, in the case of the SSM, to a solution with genuine methodological value. The final system is not a perfect representation of pleural effusion, but it is a transparent, explainable, and anatomically grounded one that demonstrates what is achievable by connecting deep learning classification, gradient-based explainability, and three-dimensional anatomical visualisation into a single coherent pipeline. The honest limitations documented throughout this dissertation map directly onto a clear agenda for future work, including radiologist evaluation of heatmap validity, broader SSM validation, and a web-based interface for educational deployment.

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Supervision Meeting Logs

The following logs document the supervision meetings held with Dr. Zenki throughout the project. They are presented in chronological order using the University of Northampton project diary format. Following the fifth formal meeting, ongoing supervision was conducted via email, with results and progress shared for approval as each phase was completed.

Meeting 1

Student:	Shazab Hassan (23851739)
Supervisor:	Dr. Zenki
Date:	October / November 2025
Last Log Date:	N/A (first meeting)

Objectives: To decide on the disease of focus, discuss the initial 3D prototype work, and establish immediate next steps for project planning and documentation.

Progress Discussed: The CheXpert dataset was reviewed and pleural effusion selected as the target condition due to its highest prevalence in the available data. An early 3D ray-casting prototype on the lung mesh was demonstrated. Discussion covered the need for landmark-based reference points on lung X-rays for proportional coordinate mapping, and two stages of image processing were identified: baseline preprocessing for CNN training and a specialised stage for coordinate extraction.

Supervisor Comments and Actions Agreed: The 3D prototype was received positively, with feedback to consider how highlighting would be made more natural once real heatmap data was available. Actions assigned: project report overview, Gantt chart, system architecture diagram, and initial image preprocessing steps.

Meeting 2

Student:	Shazab Hassan (23851739)
Supervisor:	Dr. Zenki
Date:	11 November 2025
Last Log Date:	Meeting 1

Objectives: To review datasets identified and agree on a final dataset selection strategy.

Progress Discussed: Stanford CheXpert was identified as the primary data source for its large volume, dual-view structure, and comprehensive classification labels. Additional datasets were discussed as potential supplements. Incomplete tasks from the previous week were acknowledged: the model architecture diagram and Gantt chart had not been finished.

Supervisor Comments and Actions Agreed: Dataset selection approved. Creation of working 2D and 3D mapping samples assigned as the primary task for the following week. Dataset research to be deepened including verifying prior results on CheXpert, identifying model architectures used in existing work, and examining image dimensions.

Meeting 3

Student: Shazab Hassan (23851739)
Supervisor: Dr. Zenki
Date: 25 November 2025
Last Log Date: 11 November 2025

Objectives: To review the project report structure, literature review, system architecture, and refined 3D prototype work.

Progress Discussed: Finalised dataset selection confirmed. The 3D ray-casting prototype was demonstrated with refinements. Early literature review and CNN architecture proposals were presented.

Supervisor Comments and Actions Agreed: Feedback focused on making the 3D rendering more natural and more directly connected to real heatmap data. The method of highlighting on the model needed careful thought. The supervisor set the next tasks as: completing the project report, Gantt chart, and literature review; and finalising the CNN architecture design.

Meeting 4

Student: Shazab Hassan (23851739)
Supervisor: Dr. Zenki
Date: 4 December 2025
Last Log Date: 25 November 2025

Objectives: To review four documents prepared since the last meeting: the Gantt chart, Gantt chart explanation, updated dissertation proposal (December 2025), and system workflow diagram.

Progress Discussed: All four documents were reviewed. The updated system workflow had been prepared in response to earlier discussions.

Supervisor Comments and Actions Agreed: The supervisor raised a key architectural question: why were two separate encoder networks planned? A single unified model processing both view types was suggested as simpler, more efficient, and likely equally effective. This was agreed as the approach to adopt. Consequently, Phases 2 and 4 of the Gantt chart were combined. The submission deadline was confirmed, with an agreed target to complete all development at least two weeks before final submission. Consistent progress reports using a documentation template were recommended throughout each phase.

Meeting 5

Student: Shazab Hassan (23851739)
Supervisor: Dr. Zenki
Date: 19 December 2025
Last Log Date: 4 December 2025

Objectives: To review the updated Gantt chart and system workflow incorporating the unified model approach, and to present the current literature review.

Progress Discussed: The updated system workflow was presented, now reflecting the single unified DenseNet121 processing both PA and lateral views. The literature review in progress was shown.

Supervisor Comments and Actions Agreed: The updated workflow was approved. The literature review was noted as insufficient to stand alone without Chapter 1, which had not yet been written. A deadline of Monday 22nd December was set for submission of the introduction. December goals were agreed: by the end of the month, the system should accept data input, process it, and return a classification prediction.

Note on Post-Meeting Supervision: Following Meeting 5, formal supervision meetings were replaced by email correspondence, in which completed results and phase outputs were shared with Dr. Zenki for review and approval as development progressed. This approach was agreed due to scheduling constraints in the final months of the project. Key approvals received via email included the classification results (87.24% accuracy, 94.11% AUC-ROC), the Grad-CAM++ heatmap validation results, and the 3D coordinate mapping and visualisation outputs.